

$$5. AR(2) \quad Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \varepsilon_t$$

Q. Condiciones ϕ_1, ϕ_2 causable. — $z_1 > 1$
 $z_2 > 1$

$$Y_t - \phi_1 Y_{t-1} - \phi_2 Y_{t-2} = \varepsilon_t$$

$$Y_t (1 - \phi_1 B - \phi_2 B^2) = \varepsilon_t$$

$$I(z) = 1 - \phi_1 z - \phi_2 z^2 \quad |z| > 1.$$

$$\phi_1 + \phi_2 < 1 \quad \phi_2 - \phi_1 < 1.$$

$$-\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2} > 2\phi_2$$

$$-\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2} < 2\phi_2$$

$$z = \frac{-\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{2\phi_2}$$

$$-\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2} > 2\phi_2$$

$$\pm \sqrt{\phi_1^2 + 4\phi_2} > 2\phi_2 + \phi_1$$

$$\phi_1^2 + 4\phi_2 > 4\phi_2^2 + 4\phi_2\phi_1 + \phi_1^2$$

$$4\phi_2 > \phi_2(4\phi_2 + 4\phi_1)$$

$$4 > 4\phi_2 + 4\phi_1$$

$$1 > \phi_2 + \phi_1$$

$$\phi_1^2 + 4\phi_2 > -2\phi_2 - \phi_1$$

$$\phi_1^2 + 4\phi_2 > 4\phi_2^2 - 4\phi_2\phi_1 + \phi_1^2$$

$$1 > \phi_2 - \phi_1$$

C. Relación cálculo coeficientes MA(∞)

Dado que es causable $Y_t(1 - \phi_1 B + \phi_2 B^2) = \varepsilon_t$, $Y_t = \sum_{j=0}^{\infty} \psi_j \varepsilon_{t-j}$

$$Y_t = \frac{\varepsilon_t}{1 - \phi_1 B + \phi_2 B^2} \quad \frac{1}{1 - \phi_1 B + \phi_2 B^2} = \sum_{j=0}^{\infty} \psi_j B^j \quad 1 = \sum_{j=0}^{\infty} \psi_j B^j (1 - \phi_1 B + \phi_2 B^2)$$

Multipliquemos

$$\Phi(B) Y_t = \varepsilon_t$$

Dejar Y_t solo

$\Phi(B)^{-1}$ Tiene que existir.

$$\Phi(B) \Phi(B)^{-1} = 1.$$

$$\sum_{j=0}^{\infty} \psi_j B^j - \phi_1 \sum_{j=1}^{\infty} \psi_{j-1} B^j - \phi_2 \sum_{j=2}^{\infty} \psi_{j-2} B^j = 1.$$

$$(1 - \phi_1 B - \phi_2 B^2) \left(\sum_{j=0}^{\infty} \psi_j B^j \right) = 1$$

$$\begin{aligned} \psi_0 &= 1 & \psi_1 - \phi_1 \psi_0 &= 0 \\ \psi_2 - \phi_1 \psi_1 - \phi_2 \psi_0 &= 0 \\ \psi_j - \phi_1 \psi_{j-1} - \phi_2 \psi_{j-2} &= 0 \end{aligned}$$

5.C MA(2) Invertible, relación cálculo de coeficientes AR(∞)

$$Y_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} \quad \forall t \in \mathbb{Z}, \theta_1, \theta_2 \in \mathbb{R}, \{\varepsilon_t\} \sim \text{RB}(0, \sigma^2)$$

$$Y_t = \varepsilon_t (1 + \theta_1 B + \theta_2 B^2)$$

$$(1 + \theta_1 B + \theta_2 B^2)^{-1} Y_t = \varepsilon_t \quad \text{Invertible, existe.}$$

$$(1 + \theta_1 B + \theta_2 B^2) \pi(B) = 1$$

$$\pi(B) = \frac{1}{1 + \theta_1 B + \theta_2 B^2} \quad \pi(B) = \sum_{j=0}^{\infty} \pi_j B^j$$

Coeficientes

$$(1 + \theta_1 B + \theta_2 B^2) \left(\sum_{j=0}^{\infty} \pi_j B^j \right) = 1$$

$$\pi_0 = 1 \quad \pi_1 + \theta_1 \pi_0 = 0 \quad \pi_2 + \theta_1 \pi_1 + \theta_2 \pi_0 = 0$$

$$\pi_j + \theta_1 \pi_{j-1} + \theta_2 \pi_{j-2} = 0 \quad \text{¿Cuál es su inversa?}$$

7. Considere el proceso $Y_t = \varepsilon_t - \varepsilon_{t-1}$, $\forall t \in \mathbb{Z}$; $\{\varepsilon_t\} \sim \text{RB}(0, \sigma^2)$

C. Invertible.

$$Y_t = \varepsilon_t (1 - B)$$

$$\Theta(B) = 1 - B \quad B=1 \quad \text{Tiene que estar fuera, no lo es.}$$

C. AR(∞)

$$Y_t = \varepsilon_t (1 - B) \quad \Theta(z) = (1 - z) \quad z=1.$$

$$Y_t (1 - B)^{-1} = \varepsilon_t$$

$$(1 - B)^{-1} = \sum_{j=0}^{\infty} B^j$$

Geométrica.

$$Y_t (1 + B + B^2 + B^3 + B^4) = \varepsilon_t$$

No es causable.

$$\sum_{j=0}^{\infty} B^j \rightarrow \infty \quad B=1.$$

Trabajar con ARMA

MPL.

10. Mejor predictor lineal que minimiza el error cuadrático para Y_{T+h}

$$MSE = E[(Y_{T+h} - \tilde{Y}_{T+h|T})^2] \quad \tilde{Y}_{T+h|T} = a_0 + a_1 Y_1 + \dots + a_T Y_T$$

$$\frac{d}{da_0} E[(Y_{T+h} - a_0 - \sum_{j=1}^T a_j Y_j)^2] = 0 \quad -2 E[Y_{T+h} - a_0 - \sum_{j=1}^T a_j Y_j] = 0$$

$$E[Y_{T+h}] = a_0 + \sum_{j=1}^T a_j E[Y_k Y_j], \quad k = 1, \dots, T \quad \text{Matricial}$$

$$E[\tilde{Y} \tilde{Y}^T] a = E[\tilde{Y} Y_{T+h}]$$

$$\Sigma \tilde{Y} \tilde{Y} a = \tilde{\sigma} \tilde{Y}_{T+h}$$

$$a^* = (\Sigma \tilde{Y} \tilde{Y})^{-1} \tilde{\sigma} \tilde{Y}_{T+h}$$

11. $Y_t = A \cos(Vt) + B \sin(Vt)$, $\forall t \in \mathbb{Z}$, $V \in [-\pi, \pi]$, $A, B \stackrel{\text{ind}}{\sim} N(0,1)$

MPL

a. Encuentre el mejor predictor lineal de Y_3 con base en la información de Y_1, Y_2 Como predecir Y_{t+2} dado Y_t, Y_{t+1} .

$$E[Y_t] = 0 \quad \gamma(t,s) = \text{COV}(Y_t, Y_s) = \text{COV}(V(t-s))$$

MPL de Y_3 dado Y_1, Y_2

$$E[\tilde{Y} \tilde{Y}^T] a = E[\tilde{Y} Y_3] \quad Y_{T+2|T} = -Y_T + 2 \cos V Y_{T+1}$$

$$\begin{pmatrix} \gamma(1,1) & \gamma(1,2) \\ \gamma(2,1) & \gamma(2,2) \end{pmatrix} a = \begin{pmatrix} \gamma(1,3) \\ \gamma(2,3) \end{pmatrix}$$

$$\begin{pmatrix} 1 & \cos V \\ \cos V & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} \cos 2V \\ \cos V \end{pmatrix}$$

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \frac{1}{1 - \cos^2 V} \begin{pmatrix} 1 & -\cos V \\ -\cos V & 1 \end{pmatrix} \begin{pmatrix} \cos 2V \\ \cos V \end{pmatrix}$$

$$a_1 = -1 \quad a_2 = 2 \cos V$$

$$\tilde{Y}_3 = -Y_1 + 2 \cos V Y_2$$

MA(1) Invertible.

$$Y_t = \epsilon_t + \theta \epsilon_{t-1} \quad \epsilon_t \sim WN(0, \sigma^2) \quad |\theta| < 1$$

$$\begin{pmatrix} \gamma(1,1) & \gamma(1,2) \\ \gamma(2,1) & \gamma(2,2) \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} \gamma(1,3) \\ \gamma(2,3) \end{pmatrix}$$

Construcción MPL

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \quad \bar{y}_3 = a^T y \quad a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

Minimizar

$$E[(y_3 - a^T y)^2] = 0$$

$$E[yy_3] - E[yy^T] a = 0$$

$$E[yy_3] = E[yy^T] a \quad T = E[yy^T]$$

$$T a = y \quad y = E[yy_3]$$

$$a = T^{-1} y$$

FACV

$$\gamma = \text{COV}(Y_t, Y_{t-k}) = \text{COV}(\epsilon_t + \theta \epsilon_{t-1}, \epsilon_{t-k} + \theta \epsilon_{t-k-1})$$

$$= \text{COV}(\epsilon_t, \epsilon_{t-k}) + \text{COV}(\theta \epsilon_{t-1}, \theta \epsilon_{t-k-1})$$

$$+ \text{COV}(\theta \epsilon_{t-1}, \epsilon_{t-k}) + \text{COV}(\epsilon_t, \theta \epsilon_{t-k-1})$$

$$= \begin{matrix} k=0 & \sigma^2 + \theta \sigma^2 & \sigma^2(1+\theta^2) \\ k=1 & \theta \sigma^2 & \\ k \geq 2 & 0 & \end{matrix}$$

$$\begin{pmatrix} 1+\theta^2 & \theta \\ \theta & 1+\theta^2 \end{pmatrix} \sigma^2 \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ \theta^2 \end{pmatrix} \sigma^2 \quad \begin{cases} (1+\theta^2)a_1 + \theta a_2 = 0 \\ \theta a_1 + (1+\theta^2)a_2 = \theta^2 \end{cases} \quad a_1 = \frac{-\theta^2}{1+\theta^2+\theta^4} \quad a_2 = \frac{\theta(1+\theta^2)}{1+\theta^2+\theta^4}$$

$$\tilde{Y}_3 = \frac{-\theta^2}{1+\theta^2+\theta^4} Y_1 + \frac{\theta(1+\theta^2)}{1+\theta^2+\theta^4} Y_2 \quad MSE = Y(0) - \alpha^T y = Y(0) - a_1 Y(2) - a_2 Y(1)$$

$$= \frac{\sigma^2 (1+\theta^2)(1+\theta^4)}{1+\theta^2+\theta^4}$$

AR(1) Causal.

FACV

$$Y_t = \phi_1 Y_{t-1} + \varepsilon_t$$

MA(∞)

$$Y_t - \phi_1 Y_{t-1} = \varepsilon_t$$

$$Y_t(1 - \phi_1 B) = \varepsilon_t$$

$$(1 - \phi_1 B)^{-1} = \sum_{k=0}^{\infty} (\phi_1 B)^k$$

$$Y_t = \sum_{k=0}^{\infty} (\phi_1 B)^k \varepsilon_t \quad Y_t = \sum_{k=0}^{\infty} \phi^k \varepsilon_{t-k}$$

$$\text{Cov}\left(\sum_{k=0}^{\infty} \phi^k \varepsilon_{t-k}, \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-2-j}\right)$$

$$\sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \phi^k \phi^j \text{Cov}(\varepsilon_{t-k}, \varepsilon_{t-2-j})$$

$$\sum_{j=0}^{\infty} \phi^{p+j} \phi^j \sigma^2 = \sum_{j=0}^{\infty} \phi^p \phi^{2j} \sigma^2 = \phi^p \sigma^2 \cdot \frac{1}{1-\phi^2} = \gamma(p)$$

MP

$$\bar{Y}_2 = aY_1 + bY_3$$

$$\begin{pmatrix} \gamma(0) & \gamma(2) \\ \gamma(2) & \gamma(0) \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} \gamma(1) \\ \gamma(1) \end{pmatrix}$$

$$0=b=\frac{\phi}{1+\phi^2} \quad \tilde{Y}_2 = \frac{\phi}{1+\phi^2} (Y_1 + Y_3)$$

$$MSE = E[(Y_2 - \tilde{Y}_2)^2]$$

$$MSE = Y(0) - (Y(1) \ Y(1)) T^{-1} \begin{pmatrix} \gamma(1) \\ \gamma(1) \end{pmatrix}$$

$$\text{Cov}(Y_t, Y_{t-k}) = \text{Cov}(\phi_1 Y_{t-1} + \varepsilon_t, \phi_1 Y_{t-k-1} + \varepsilon_{t-k})$$

$$= \text{Cov}(\phi_1 Y_{t-1}, \phi_1 Y_{t-k-1}) + \text{Cov}(\varepsilon_t, \varepsilon_{t-k})$$

$$+ \text{Cov}(\phi_1 Y_{t-1}, \varepsilon_{t-k}) + \text{Cov}(\varepsilon_t, \phi_1 Y_{t-k-1})$$

$$Y_0 = \phi^2 Y_0 + \sigma^2 \quad Y_1 = \phi_1 \text{Cov}(Y_{t-1}, \varepsilon_{t+1})$$

$$Y_0 = \frac{\sigma^2}{1-\phi^2} \quad Y_1 = \phi_1 Y_0$$

$$Y_2 = \text{Cov}(Y_t, Y_{t-2}) = \text{Cov}(\phi Y_{t-1} + \varepsilon_t, Y_{t-2})$$

$$Y_2 = \phi \gamma(1) = \phi \text{Cov}(Y_t, Y_{t-1}) = \phi \text{Cov}(\phi Y_{t-1} + \varepsilon_t, Y_{t-1})$$

$$Y_2 = \phi (\phi \gamma(0)) = \phi^2 \cdot \frac{\sigma^2}{1-\phi^2}$$

$$\gamma(0) \begin{pmatrix} 1 & \phi^2 \\ \phi^2 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \gamma(0) \begin{pmatrix} \phi \\ \phi \end{pmatrix}$$

$$a + \phi^2 b = \phi \quad \phi^2 a + b = \phi$$

$$MSE = Y(0) - \alpha^T y = Y(0) - a_1 Y(2) - a_2 Y(1)$$

$$= Y(0) - 2a\gamma(1) = Y(0) - 2\left(\frac{\phi}{1+\phi^2}\right) \cdot \gamma(0) \phi$$

$$= \frac{\sigma^2}{1-\phi^2} \cdot \left(\frac{1-\phi^2}{1+\phi^2}\right) = \frac{\sigma^2}{1+\phi^2}$$

Para un arima(0,1,0), el modelo es:

$$Y_t = Y_{t-1} + \varepsilon_t \quad \varepsilon_t \sim (0, \sigma^2)$$

FACV

$$\text{COV}(Y_t, Y_{t-k}) = \text{COV}(Y_{t-1} + \varepsilon_t, Y_{t-k-1} + \varepsilon_{t-k})$$

$$\text{COV}(Y_{t-1}, Y_{t-k-1}) + \text{COV}(\varepsilon_t, \varepsilon_{t-k}) + \text{COV}(Y_{t-1}, \varepsilon_{t-k}) + \text{COV}(\varepsilon_t, Y_{t-k-1})$$

$$t=2 \quad k=1 \quad \text{COV}(Y_2, Y_1) = \sigma^2$$

$$t=3 \quad k=1 \quad \text{COV}(Y_3, Y_2) = \sigma^2 + \sigma^2 = 2\sigma^2$$

$$t=3 \quad k=2 \quad \text{COV}(Y_3, Y_1) = \sigma^2$$

$$Y_t = Y_0 + \sum_{i=1}^t \varepsilon_i \quad \text{Var}(Y_0) + \sum \text{Var}(\varepsilon_i)$$

$$\text{COV}(Y_2, \varepsilon_1)$$

$$\text{COV}(Y_1 + \varepsilon_2, \varepsilon_1)$$

$$\text{COV}(Y_1, \varepsilon_1) = \text{COV}(Y_0 + \varepsilon_1, \varepsilon_1) = \sigma^2$$

MP2:

$$\text{COV}(Y_2, Y_1) = a \text{Var}(Y_1) + b \text{COV}(Y_3, Y_1)$$

$$\text{COV}(Y_1, Y_3) = a \text{COV}(Y_1, Y_3) + b \text{Var}(Y_3)$$

$$\sigma^2 = a\sigma^2 + b\sigma^2 \rightarrow a + b = 1$$

$$2\sigma^2 = a\sigma^2 + 3b\sigma^2 \rightarrow a + 3b = 2$$

$$a = b = \frac{1}{2} \rightarrow \tilde{Y}_2 = \frac{1}{2} Y_1 + \frac{1}{2} Y_3$$

$$Y_i = \frac{1}{2} Y_{i-1} + \frac{1}{2} Y_{i+1} \quad \text{Promedio.}$$

$$\text{MSE} = \text{Var}(e_i) = \text{Var}\left(\frac{1}{2} e_i - \frac{1}{2} e_{i+1}\right) = \frac{1}{4} \text{Var}(e_i) + \frac{1}{4} \text{Var}(e_{i+1}) = \frac{1}{4} \sigma^2 + \frac{1}{4} \sigma^2 = \frac{\sigma^2}{2}$$

